

Intro to Machine Learning: Linear and Polynomial Regression

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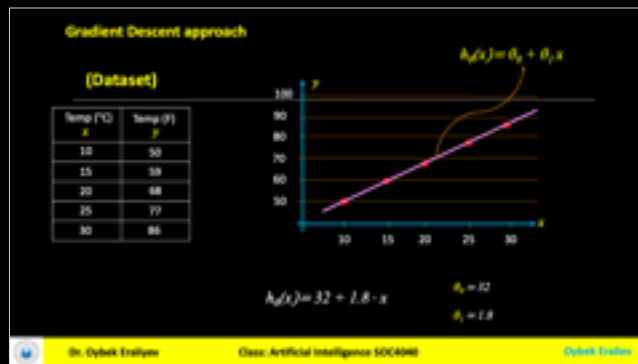


Content

- **Linear Regression with Multi-Variables**
- **Polynomial Regression**
- **Setting Up Environments**
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Approaches for solving Linear regression problems:

Gradient Descent



Normal Equation

Linear Regression:

Normal Equation

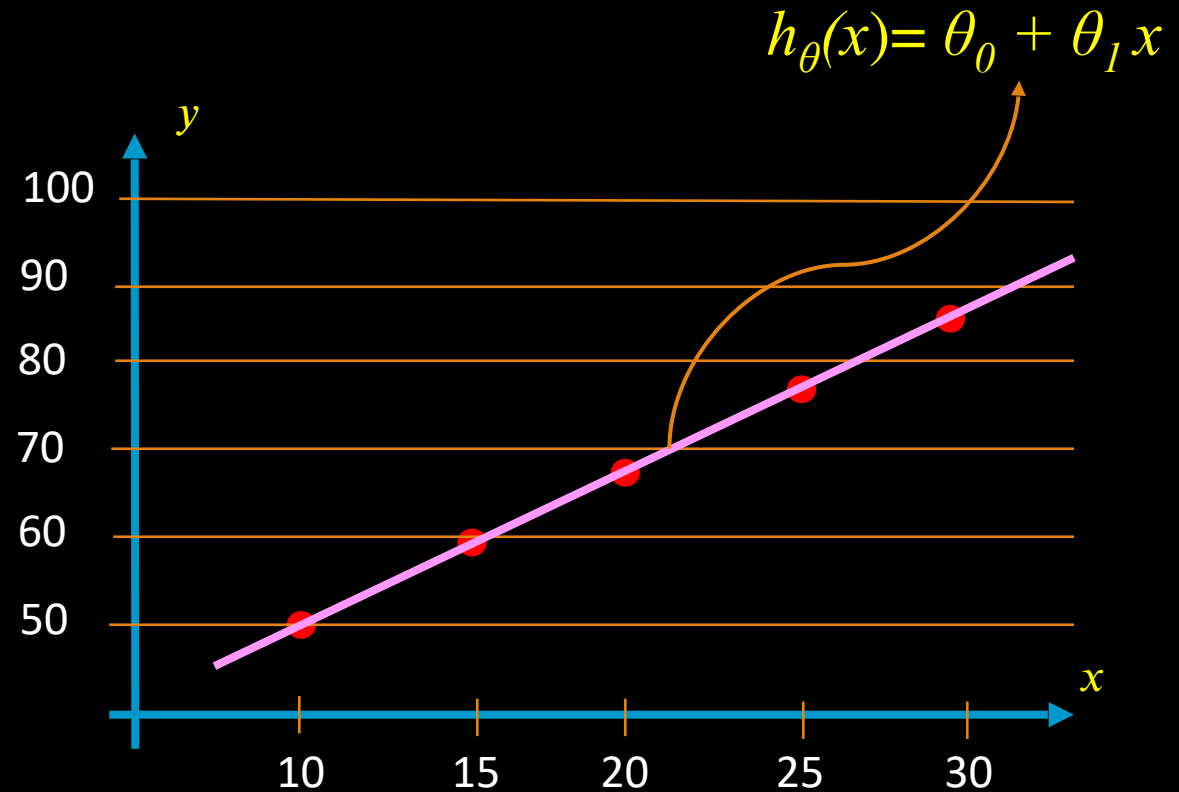
$$\theta = (X^T \cdot X)^{-1} \cdot (X^T \cdot y)$$

Dr. Oybek Eraliyev Class: Artificial Intelligence SOC4040 Oybek Eraliev

Gradient Descent approach

(Dataset)

Temp (°C) x	Temp (F) y
10	50
15	59
20	68
25	77
30	86



$$h_{\theta}(x) = 32 + 1.8 \cdot x$$

$$\theta_0 = 32$$

$$\theta_1 = 1.8$$

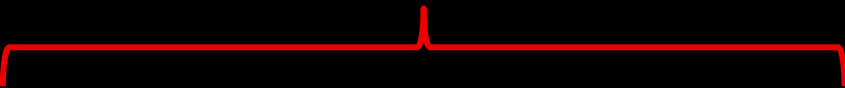
Gradient Descent approach

(Dataset)

Temp (°C) x	Temp (F) y
10	50
15	59
20	68
25	77
30	86

Gradient Descent approach

(Dataset)
 n



x_1	x_2	x_3	y
10	12	5	50
15	11	7	59
20	23	3	68
25	28	6	77
30	31	6	86

Hypothesis

$$h(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3$$



$$h(x) = \theta_0 + \sum_{i=1}^n \theta_i x_i$$

Gradient Descent approach

(Dataset)

x_1	x_2	x_3	y
10	12	5	50
15	11	7	59
20	23	3	68
25	28	6	77
30	31	6	86

Hypothesis

$$h(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3$$

Loss function

$$J(\theta) = \frac{1}{2m} \sum_{i=1}^m \left(h_{\theta}(x^{(i)}) - y^{(i)} \right)^2$$

Minimize Loss

$$\underset{\theta}{\text{minimize}} \quad J(\theta)$$

Gradient descent algorithm

$$\theta_j = \theta_j - \alpha \cdot \frac{\partial}{\partial \theta_j} J(\theta)$$

Gradient Descent approach

(Dataset)

x_1	x_2	x_3	y
10	12	5	50
15	11	7	59
20	23	3	68
25	28	6	77
30	31	6	86

Hypothesis

$$h(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3$$

Gradient Descent approach

(Dataset)

x_1	x_2	x_3	y
10	12	5	50
15	11	7	59
20	23	3	68
25	28	6	77
30	31	6	86

Hypothesis

$$h(x) = \theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3$$

Gradient Descent approach

(Dataset)

X				y
x_0	x_1	x_2	x_3	y
1	10	12	5	50
1	15	11	7	59
1	20	23	3	68
1	25	28	6	77
1	30	31	6	86

Hypothesis

$$h(x) = \theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3$$



$$h(x) = X\theta$$

Gradient Descent approach

Hypothesis

$$h(x) = \theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3$$

$$X = \begin{bmatrix} 1 & 10 & 12 & 5 \\ 1 & 15 & 11 & 7 \\ 1 & 20 & 23 & 3 \\ 1 & 25 & 28 & 6 \\ 1 & 30 & 31 & 6 \end{bmatrix} \quad \theta = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad y = \begin{bmatrix} 50 \\ 59 \\ 68 \\ 77 \\ 86 \end{bmatrix}$$


$$h(x) = X\theta$$

Gradient Descent approach

Hypothesis

$$\boxed{h(x) = X\theta} = \begin{bmatrix} 1 & 10 & 12 & 5 \\ 1 & 15 & 11 & 7 \\ 1 & 20 & 23 & 3 \\ 1 & 25 & 28 & 6 \\ 1 & 30 & 31 & 6 \end{bmatrix} \times \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} =$$
$$y = \begin{bmatrix} 50 \\ 59 \\ 68 \\ 77 \\ 86 \end{bmatrix}$$



Gradient Descent approach

Hypothesis

$$\boxed{h(x) = X\theta} = \begin{bmatrix} 1 & 10 & 12 & 5 \\ 1 & 15 & 11 & 7 \\ 1 & 20 & 23 & 3 \\ 1 & 25 & 28 & 6 \\ 1 & 30 & 31 & 6 \end{bmatrix} \times \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \times 0 + 10 \times 0 + 12 \times 0 + 5 \times 0 \\ \vdots \end{bmatrix}$$
$$y = \begin{bmatrix} 50 \\ 59 \\ 68 \\ 77 \\ 86 \end{bmatrix}$$

Gradient Descent approach

Hypothesis

$$\boxed{h(x) = X\theta} = \begin{bmatrix} 1 & 10 & 12 & 5 \\ 1 & 15 & 11 & 7 \\ 1 & 20 & 23 & 3 \\ 1 & 25 & 28 & 6 \\ 1 & 30 & 31 & 6 \end{bmatrix} \times \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \times 0 + 10 \times 0 + 12 \times 0 + 5 \times 0 \\ 1 \times 0 + 15 \times 0 + 11 \times 0 + 7 \times 0 \\ \\ \\ \end{bmatrix}$$
$$y = \begin{bmatrix} 50 \\ 59 \\ 68 \\ 77 \\ 86 \end{bmatrix}$$

Gradient Descent approach

Hypothesis

$$\boxed{h(x) = X\theta} = \begin{bmatrix} 1 & 10 & 12 & 5 \\ 1 & 15 & 11 & 7 \\ 1 & 20 & 23 & 3 \\ 1 & 25 & 28 & 6 \\ 1 & 30 & 31 & 6 \end{bmatrix} \times \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \times 0 + 10 \times 0 + 12 \times 0 + 5 \times 0 \\ 1 \times 0 + 15 \times 0 + 11 \times 0 + 7 \times 0 \\ 1 \times 0 + 20 \times 0 + 23 \times 0 + 3 \times 0 \end{bmatrix}$$
$$y = \begin{bmatrix} 50 \\ 59 \\ 68 \\ 77 \\ 86 \end{bmatrix}$$

Gradient Descent approach

Hypothesis

$$\boxed{h(x) = X\theta} = \begin{bmatrix} 1 & 10 & 12 & 5 \\ 1 & 15 & 11 & 7 \\ 1 & 20 & 23 & 3 \\ 1 & 25 & 28 & 6 \\ 1 & 30 & 31 & 6 \end{bmatrix} \times \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \times 0 + 10 \times 0 + 12 \times 0 + 5 \times 0 \\ 1 \times 0 + 15 \times 0 + 11 \times 0 + 7 \times 0 \\ 1 \times 0 + 20 \times 0 + 23 \times 0 + 3 \times 0 \\ 1 \times 0 + 25 \times 0 + 28 \times 0 + 6 \times 0 \end{bmatrix}$$
$$y = \begin{bmatrix} 50 \\ 59 \\ 68 \\ 77 \\ 86 \end{bmatrix}$$

Gradient Descent approach

Hypothesis

$$\boxed{h(x) = X\theta} = \begin{bmatrix} 1 & 10 & 12 & 5 \\ 1 & 15 & 11 & 7 \\ 1 & 20 & 23 & 3 \\ 1 & 25 & 28 & 6 \\ 1 & 30 & 31 & 6 \end{bmatrix} \times \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \times 0 + 10 \times 0 + 12 \times 0 + 5 \times 0 \\ 1 \times 0 + 15 \times 0 + 11 \times 0 + 7 \times 0 \\ 1 \times 0 + 20 \times 0 + 23 \times 0 + 3 \times 0 \\ 1 \times 0 + 25 \times 0 + 28 \times 0 + 6 \times 0 \\ 1 \times 0 + 30 \times 0 + 31 \times 0 + 6 \times 0 \end{bmatrix}$$
$$y = \begin{bmatrix} 50 \\ 59 \\ 68 \\ 77 \\ 86 \end{bmatrix}$$

Gradient Descent approach

Hypothesis

Initial prediction

$$\boxed{h(x) = X\theta} = \begin{bmatrix} 1 & 10 & 12 & 5 \\ 1 & 15 & 11 & 7 \\ 1 & 20 & 23 & 3 \\ 1 & 25 & 28 & 6 \\ 1 & 30 & 31 & 6 \end{bmatrix} \times \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \times 0 + 10 \times 0 + 12 \times 0 + 5 \times 0 \\ 1 \times 0 + 15 \times 0 + 11 \times 0 + 7 \times 0 \\ 1 \times 0 + 20 \times 0 + 23 \times 0 + 3 \times 0 \\ 1 \times 0 + 25 \times 0 + 28 \times 0 + 6 \times 0 \\ 1 \times 0 + 30 \times 0 + 31 \times 0 + 6 \times 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$y = \begin{bmatrix} 50 \\ 59 \\ 68 \\ 77 \\ 86 \end{bmatrix}$

Gradient Descent approach

Hypothesis

$$h(x) = X\theta = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$y = \begin{bmatrix} 50 \\ 59 \\ 68 \\ 77 \\ 86 \end{bmatrix}$$

$$(h(x) - y) = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} - \begin{bmatrix} 50 \\ 59 \\ 68 \\ 77 \\ 86 \end{bmatrix} = \begin{bmatrix} -50 \\ -59 \\ -68 \\ -77 \\ -86 \end{bmatrix}$$

Loss function

$$J(\theta) = \frac{1}{2m} \sum_{i=1}^m \left(h_{\theta}(x^{(i)}) - y^{(i)} \right)^2$$

$$J(\theta) = \frac{1}{2 \times 5} \sum_{i=1}^5 (h(x) - y)^2 =$$

$$= \frac{1}{2 \times 5} \left((-50)^2 + (-59)^2 + (-68)^2 + (-77)^2 + (-86)^2 \right)$$

$$= \frac{1}{2 \times 5} (23930) = 2393$$

Gradient Descent approach

Hypothesis

$$h(x) = X\theta = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad y = \begin{bmatrix} 50 \\ 59 \\ 68 \\ 77 \\ 86 \end{bmatrix} \quad \theta = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Loss function

$$J(\theta) = 2393$$

Minimize Loss

$$\underset{\theta}{\text{minimize}} \quad J(\theta)$$

Gradient descent

$$\theta_j = \theta_j - \alpha \cdot \frac{\partial}{\partial \theta_j} J(\theta) \quad \alpha = 0.001$$

$$\theta = \theta^0 - \alpha \cdot \frac{1}{m} \sum_{i=1}^m \left(h_{\theta}(x^{(i)}) - y^{(i)} \right) \cdot x^{(i)} =$$

$$= \theta^0 - \alpha \cdot \frac{1}{m} \cdot X^T \cdot (h(x) - y)$$

$$\theta = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} - \alpha \cdot \frac{1}{5} \cdot \begin{bmatrix} 1 & 10 & 12 & 5 \\ 1 & 15 & 11 & 7 \\ 1 & 20 & 23 & 3 \\ 1 & 25 & 28 & 6 \\ 1 & 30 & 31 & 6 \end{bmatrix}^T \cdot \left[\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} - \begin{bmatrix} 50 \\ 59 \\ 68 \\ 77 \\ 86 \end{bmatrix} \right]$$

Gradient Descent approach

Hypothesis

$$h(x) = X\theta = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad y = \begin{bmatrix} 50 \\ 59 \\ 68 \\ 77 \\ 86 \end{bmatrix} \quad \theta = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Loss function

$$J(\theta) = 2393$$

Minimize Loss

$$\underset{\theta}{\text{minimize}} \quad J(\theta)$$

Gradient descent

$$\theta = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} - \alpha \cdot \frac{1}{5} \cdot \begin{bmatrix} 1 & 10 & 12 & 5 \\ 1 & 15 & 11 & 7 \\ 1 & 20 & 23 & 3 \\ 1 & 25 & 28 & 6 \\ 1 & 30 & 31 & 6 \end{bmatrix}^T \cdot \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} - \begin{bmatrix} 50 \\ 59 \\ 68 \\ 77 \\ 86 \end{bmatrix}$$

Gradient Descent approach

Hypothesis

$$h(x) = X\theta = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad y = \begin{bmatrix} 50 \\ 59 \\ 68 \\ 77 \\ 86 \end{bmatrix} \quad \theta = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Loss function

$$J(\theta) = 2393$$

Minimize Loss

$$\underset{\theta}{\text{minimize}} \quad J(\theta)$$

Gradient descent

$$\theta = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} - \alpha \cdot \frac{1}{5} \cdot \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 10 & 15 & 20 & 25 & 30 \\ 12 & 11 & 23 & 28 & 31 \\ 5 & 7 & 3 & 6 & 6 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} - \begin{bmatrix} 50 \\ 59 \\ 68 \\ 77 \\ 86 \end{bmatrix}$$

Gradient Descent approach

Hypothesis

$$h(x) = X\theta = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} y = \begin{bmatrix} 50 \\ 59 \\ 68 \\ 77 \\ 86 \end{bmatrix} \theta = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Loss function

$$J(\theta) = 2393$$

Minimize Loss

$$\underset{\theta}{\text{minimize}} \quad J(\theta)$$

$$\alpha = 0.001$$

Gradient descent

$$\theta = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} - \alpha \cdot \frac{1}{5} \cdot \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 10 & 15 & 20 & 25 & 30 \\ 12 & 11 & 23 & 28 & 31 \\ 5 & 7 & 3 & 6 & 6 \end{bmatrix} \cdot \begin{bmatrix} -50 \\ -59 \\ -68 \\ -77 \\ -86 \end{bmatrix} =$$

$$= \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} - \alpha \cdot \frac{1}{5} \cdot \begin{bmatrix} -340 \\ -7250 \\ -7635 \\ -1845 \end{bmatrix} = \begin{bmatrix} 0.06 \\ 1.45 \\ 1.527 \\ 0.369 \end{bmatrix}$$

Gradient Descent approach

Hypothesis

$$h(x) = X\theta \quad y = \begin{bmatrix} 50 \\ 59 \\ 68 \\ 77 \\ 86 \end{bmatrix} \quad \theta = \begin{bmatrix} 0.06 \\ 1.45 \\ 1.527 \\ 0.369 \end{bmatrix}$$

Loss function

$$J(\theta) = 2393 \rightarrow \text{in the first iteration}$$

Minimize Loss

$$\underset{\theta}{\text{minimize}} \quad J(\theta)$$

$$\alpha = 0.001$$

Loss function

$$J(\theta) = \frac{1}{2 \times 5} \sum_{i=1}^5 (h(x) - y)^2$$

Gradient Descent approach

Hypothesis

$$h(x) = X\theta \quad y = \begin{bmatrix} 50 \\ 59 \\ 68 \\ 77 \\ 86 \end{bmatrix} \quad \theta = \begin{bmatrix} 0.06 \\ 1.45 \\ 1.527 \\ 0.369 \end{bmatrix}$$

Loss function

$$J(\theta) = 2393 \rightarrow \text{in the first iteration}$$

Minimize Loss

$$\underset{\theta}{\text{minimize}} \quad J(\theta)$$

$$\alpha = 0.001$$

Loss function

$$J(\theta) = \frac{1}{2 \times 5} \sum_{i=1}^5 (h(x) - y)^2$$

$$h(x) = X\theta = \begin{bmatrix} 1 & 10 & 12 & 5 \\ 1 & 15 & 11 & 7 \\ 1 & 20 & 23 & 3 \\ 1 & 25 & 28 & 6 \\ 1 & 30 & 31 & 6 \end{bmatrix} \cdot$$

Gradient Descent approach

Hypothesis

$$h(x) = X\theta \quad y = \begin{bmatrix} 50 \\ 59 \\ 68 \\ 77 \\ 86 \end{bmatrix} \quad \theta = \begin{bmatrix} 0.06 \\ 1.45 \\ 1.527 \\ 0.369 \end{bmatrix}$$

Loss function

$$J(\theta) = 2393 \rightarrow \text{in the first iteration}$$

Minimize Loss

$$\underset{\theta}{\text{minimize}} \quad J(\theta)$$

$$\alpha = 0.001$$

Loss function

$$J(\theta) = \frac{1}{2 \times 5} \sum_{i=1}^5 (h(x) - y)^2$$

$$h(x) = X\theta = \begin{bmatrix} 1 & 10 & 12 & 5 \\ 1 & 15 & 11 & 7 \\ 1 & 20 & 23 & 3 \\ 1 & 25 & 28 & 6 \\ 1 & 30 & 31 & 6 \end{bmatrix} \cdot \begin{bmatrix} 0.06 \\ 1.45 \\ 1.527 \\ 0.369 \end{bmatrix} = \begin{bmatrix} 34.737 \\ 41.198 \\ 65.296 \\ 81.288 \\ 93.119 \end{bmatrix}$$

Gradient Descent approach

Hypothesis

$$h(x) = X\theta \quad y = \begin{bmatrix} 50 \\ 59 \\ 68 \\ 77 \\ 86 \end{bmatrix} \quad \theta = \begin{bmatrix} 0.06 \\ 1.45 \\ 1.527 \\ 0.369 \end{bmatrix}$$

Loss function

$$J(\theta) = 2393 \rightarrow \text{in the first iteration}$$

Minimize Loss

$$\underset{\theta}{\text{minimize}} \quad J(\theta)$$

$$\alpha = 0.001$$

Loss function

$$J(\theta) = \frac{1}{2 \times 5} \sum_{i=1}^5 (h(x) - y)^2$$

$$(h(x) - y) = \begin{bmatrix} 34.737 \\ 41.198 \\ 65.296 \\ 81.288 \\ 93.119 \end{bmatrix} -$$

Gradient Descent approach

Hypothesis

$$h(x) = X\theta \quad y = \begin{bmatrix} 50 \\ 59 \\ 68 \\ 77 \\ 86 \end{bmatrix} \quad \theta = \begin{bmatrix} 0.06 \\ 1.45 \\ 1.527 \\ 0.369 \end{bmatrix}$$

Loss function

$$J(\theta) = 2393 \rightarrow \text{in the first iteration}$$

Minimize Loss

$$\underset{\theta}{\text{minimize}} \quad J(\theta)$$

$$\alpha = 0.001$$

Loss function

$$J(\theta) = \frac{1}{2 \times 5} \sum_{i=1}^5 (h(x) - y)^2$$

$$(h(x) - y) = \begin{bmatrix} 34.737 \\ 41.198 \\ 65.296 \\ 81.288 \\ 93.119 \end{bmatrix} - \begin{bmatrix} 50 \\ 59 \\ 68 \\ 77 \\ 86 \end{bmatrix} = \begin{bmatrix} -15.263 \\ -17.802 \\ -2.704 \\ 4.288 \\ 7.119 \end{bmatrix}$$

Gradient Descent approach

Hypothesis

$$h(x) = X\theta \quad y = \begin{bmatrix} 50 \\ 59 \\ 68 \\ 77 \\ 86 \end{bmatrix} \quad \theta = \begin{bmatrix} 0.06 \\ 1.45 \\ 1.527 \\ 0.369 \end{bmatrix}$$

Loss function

$$J(\theta) = 2393 \rightarrow \text{in the first iteration}$$

Minimize Loss

$$\underset{\theta}{\text{minimize}} \quad J(\theta)$$

$$\alpha = 0.001$$

Loss function

$$J(\theta) = \frac{1}{2 \times 5} \sum_{i=1}^5 (h(x) - y)^2 = \frac{1}{2 \times 5}$$

$$\left((-15.263)^2 + (-17.802)^2 + (-2.704)^2 + (4.288)^2 + (7.119)^2 \right)$$

$$= \frac{1}{2 \times 5} (626.2491) = 62.625$$

Gradient Descent approach

Hypothesis

$$h(x) = X\theta$$

Loss function

$$J(\theta) = \frac{1}{2m} \sum_{i=1}^m \left(h_{\theta}(x^{(i)}) - y^{(i)} \right)^2$$

Minimize Loss

$$\underset{\theta}{\text{minimize}} \quad J(\theta)$$

Gradient descent

$$\theta_j = \theta_j - \alpha \cdot \frac{\partial}{\partial \theta_j} J(\theta)$$

$$\alpha = 0.001$$

$$X = \begin{bmatrix} 1 & 10 & 12 & 5 \\ 1 & 15 & 11 & 7 \\ 1 & 20 & 23 & 3 \\ 1 & 25 & 28 & 6 \\ 1 & 30 & 31 & 6 \end{bmatrix} \quad y = \begin{bmatrix} 50 \\ 59 \\ 68 \\ 77 \\ 86 \end{bmatrix}$$

Iteration 1

$$\theta = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad h(x) = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad J(\theta) = 2393$$

Iteration 2

$$\theta = \begin{bmatrix} 0.06 \\ 1.45 \\ 1.527 \\ 0.369 \end{bmatrix} \quad h(x) = \begin{bmatrix} 34.737 \\ 41.198 \\ 65.296 \\ 81.288 \\ 93.119 \end{bmatrix} \quad J(\theta) = 62.625$$

Iteration 3

$$\theta = \begin{bmatrix} 0.073 \\ 1.481 \\ 1.547 \\ 0.397 \end{bmatrix} \quad h(x) = \begin{bmatrix} 35.429 \\ 42.08 \\ 66.459 \\ 82.789 \\ 94.833 \end{bmatrix} \quad J(\theta) = 61.25$$

Python Code: Linear Regression

Gradient Descent Approach

```
import matplotlib.pyplot as plt
import numpy as np
```

[2] ✓ 0.0s

```
dataset = np.array([[10, 12, 5, 50],
                    [15, 11, 7, 59],
                    [20, 23, 3, 68],
                    [25, 28, 6, 77],
                    [30, 31, 6, 86]])
```

```
print(dataset.shape)
```

[3] ✓ 0.0s

... (5, 4)

```
X = dataset[:, 0:3]
y = dataset[:, -1]

y = y.reshape(len(y), 1)

print(X.shape)
print(y.shape)
```

[6] ✓ 0.0s

... (5, 3)
(5, 1)

```
x0 = np.ones((5,1))
X_fin = np.hstack((x0, X))
print(X_fin.shape)
print(X_fin)
```

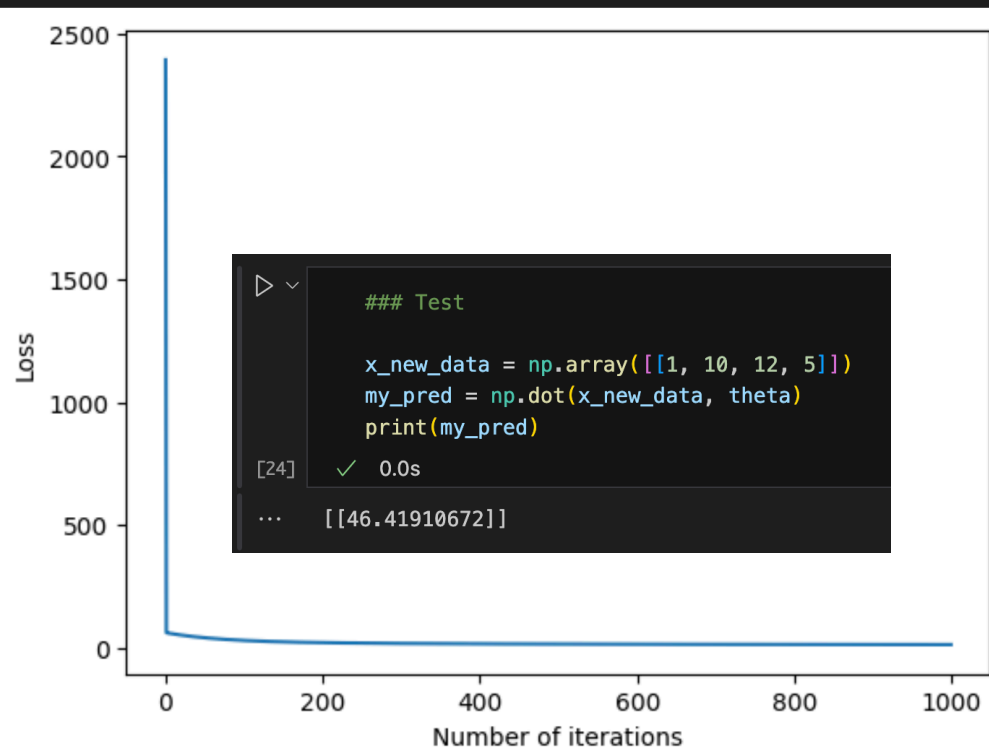
[10] ✓ 0.0s

... (5, 4)
[[1. 10. 12. 5.]
 [1. 15. 11. 7.]
 [1. 20. 23. 3.]
 [1. 25. 28. 6.]
 [1. 30. 31. 6.]]

Python Code: Linear Regression

```
x_axis = np.array(range(iteration))
y_axis = np.array(Loss)
plt.plot(x_axis, y_axis)
plt.xlabel("Number of iterations")
plt.ylabel("Loss")
plt.show()
```

[23] ✓ 0.0s



```
learning_rate = 0.001
iteration = 1000
Loss = []
```

```
theta = np.zeros((4,1))
m = len(X_fin)
```

```
for iter in range(iteration):
    h_x = np.dot(X_fin, theta)
    error = h_x - y
    loss = np.sum(error**2)/(2*m)
    Loss.append(loss)
    delta = np.dot(np.transpose(X_fin), error)
    theta = theta - learning_rate*delta/m
    if iter%100 == 0:
        print(f"{iter} iteration, Loss = {loss}")
```

[22] ✓ 0.0s

```
... 0 iteration, Loss = 2393.0
100 iteration, Loss = 30.521344198917877
200 iteration, Loss = 21.534903639951743
300 iteration, Loss = 18.402239983851256
400 iteration, Loss = 16.906975623825854
500 iteration, Loss = 15.97142590761768
600 iteration, Loss = 15.294812630399145
700 iteration, Loss = 14.77477744809284
800 iteration, Loss = 14.364228699680003
900 iteration, Loss = 14.034808226796699
```

Linear Regression:

Normal Equation

$$\theta = (X^T \cdot X)^{-1} \cdot (X^T \cdot y)$$

Linear Regression: Normal Equation

(Dataset)

X				y
x_0	x_1	x_2	x_3	y
1	10	12	5	50
1	15	11	7	59
1	20	23	3	68
1	25	28	6	77
1	30	31	6	86

Hypothesis

$$h(x) = X\theta$$

$$\theta = (X^T \cdot X)^{-1} \cdot (X^T \cdot y)$$

Linear Regression: Normal Equation

(Dataset)

$$X = \begin{bmatrix} 1 & 10 & 12 & 5 \\ 1 & 15 & 11 & 7 \\ 1 & 20 & 23 & 3 \\ 1 & 25 & 28 & 6 \\ 1 & 30 & 31 & 6 \end{bmatrix} \quad y = \begin{bmatrix} 50 \\ 59 \\ 68 \\ 77 \\ 86 \end{bmatrix}$$

Hypothesis

$$h(x) = X\theta$$

$$\theta = (X^T \cdot X)^{-1} \cdot (X^T \cdot y)$$

Linear Regression: Normal Equation

$$X = \begin{bmatrix} 1 & 10 & 12 & 5 \\ 1 & 15 & 11 & 7 \\ 1 & 20 & 23 & 3 \\ 1 & 25 & 28 & 6 \\ 1 & 30 & 31 & 6 \end{bmatrix} \quad y = \begin{bmatrix} 50 \\ 59 \\ 68 \\ 77 \\ 86 \end{bmatrix}$$

$$\theta = (X^T \cdot X)^{-1} \cdot (X^T \cdot y)$$

$$X^T =$$

Linear Regression: Normal Equation

$$X = \begin{bmatrix} 1 & 10 & 12 & 5 \\ 1 & 15 & 11 & 7 \\ 1 & 20 & 23 & 3 \\ 1 & 25 & 28 & 6 \\ 1 & 30 & 31 & 6 \end{bmatrix} \quad y = \begin{bmatrix} 50 \\ 59 \\ 68 \\ 77 \\ 86 \end{bmatrix}$$

$$X^T = \begin{bmatrix} 1 & 10 & 12 & 5 \\ 1 & 15 & 11 & 7 \\ 1 & 20 & 23 & 3 \\ 1 & 25 & 28 & 6 \\ 1 & 30 & 31 & 6 \end{bmatrix}^T$$

$$\theta = (X^T \cdot X)^{-1} \cdot (X^T \cdot y)$$

Linear Regression: Normal Equation

$$X = \begin{bmatrix} 1 & 10 & 12 & 5 \\ 1 & 15 & 11 & 7 \\ 1 & 20 & 23 & 3 \\ 1 & 25 & 28 & 6 \\ 1 & 30 & 31 & 6 \end{bmatrix} \quad y = \begin{bmatrix} 50 \\ 59 \\ 68 \\ 77 \\ 86 \end{bmatrix}$$

$$X^T = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 10 & 15 & 20 & 25 & 30 \\ 12 & 11 & 23 & 28 & 31 \\ 5 & 7 & 3 & 6 & 6 \end{bmatrix}$$

$$\theta = (X^T \cdot X)^{-1} \cdot (X^T \cdot y)$$

Linear Regression: Normal Equation

$$\theta = (X^T \cdot X)^{-1} \cdot (X^T \cdot y)$$

$$\theta = \underbrace{\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 10 & 15 & 20 & 25 & 30 \\ 12 & 11 & 23 & 28 & 31 \\ 5 & 7 & 3 & 6 & 6 \end{bmatrix} \cdot \begin{bmatrix} 1 & 10 & 12 & 5 \\ 1 & 15 & 11 & 7 \\ 1 & 20 & 23 & 3 \\ 1 & 25 & 28 & 6 \\ 1 & 30 & 31 & 6 \end{bmatrix}}^{-1} \cdot \underbrace{\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 10 & 15 & 20 & 25 & 30 \\ 12 & 11 & 23 & 28 & 31 \\ 5 & 7 & 3 & 6 & 6 \end{bmatrix} \cdot \begin{bmatrix} 50 \\ 59 \\ 68 \\ 77 \\ 86 \end{bmatrix}}$$

Linear Regression: Normal Equation

$$\theta = (X^T \cdot X)^{-1} \cdot (X^T \cdot y)$$

$$\theta = \begin{bmatrix} 5 & 100 & 105 & 27 \\ 100 & 2250 & 2375 & 545 \\ 105 & 2375 & 2539 & 560 \\ 27 & 545 & 560 & 155 \end{bmatrix}^{-1} \cdot \begin{bmatrix} 340 \\ 7250 \\ 7635 \\ 1845 \end{bmatrix}$$

Linear Regression: Normal Equation

$$\theta = (X^T \cdot X)^{-1} \cdot (X^T \cdot y)$$

$$\theta = \begin{bmatrix} 6.95 & 0.39 & -0.41 & -1.11 \\ 0.39 & 0.09 & -0.08 & -0.11 \\ -0.41 & -0.08 & 0.07 & 0.1 \\ -1.11 & -0.01 & 0.1 & 0.24 \end{bmatrix} \cdot \begin{bmatrix} 340 \\ 7250 \\ 7635 \\ 1845 \end{bmatrix} = \begin{bmatrix} 32 \\ 1.8 \\ 2.8e(-14) \\ -1.7e(-13) \end{bmatrix}$$

Linear Regression:

Gradient descent

- ✓ need to choose α learning rate
- ✓ To find optimal solution a lot of iterations are required.
- ✓ Very good, if the number of features is very large.

Normal Equation

- ✓ No need to choose α learning rate
- ✓ no need to iterations
- ✓ Very slow, if the number of features is very large.
- ✓ $(X^T \cdot X)^{-1}$ need to be calculated.

Python Code: Linear Regression

```
theta_norm = np.linalg.inv(X_fin.T.dot(X_fin)).dot(X_fin.T).dot(y)
print(theta_norm)
```

✓ 0.0s

```
[[ 3.20000000e+01]
 [ 1.80000000e+00]
 [-3.33066907e-14]
 [-2.58681965e-14]]
```

```
my_nor_pred = np.dot(x_new_data, theta_norm)
print(my_nor_pred)
```

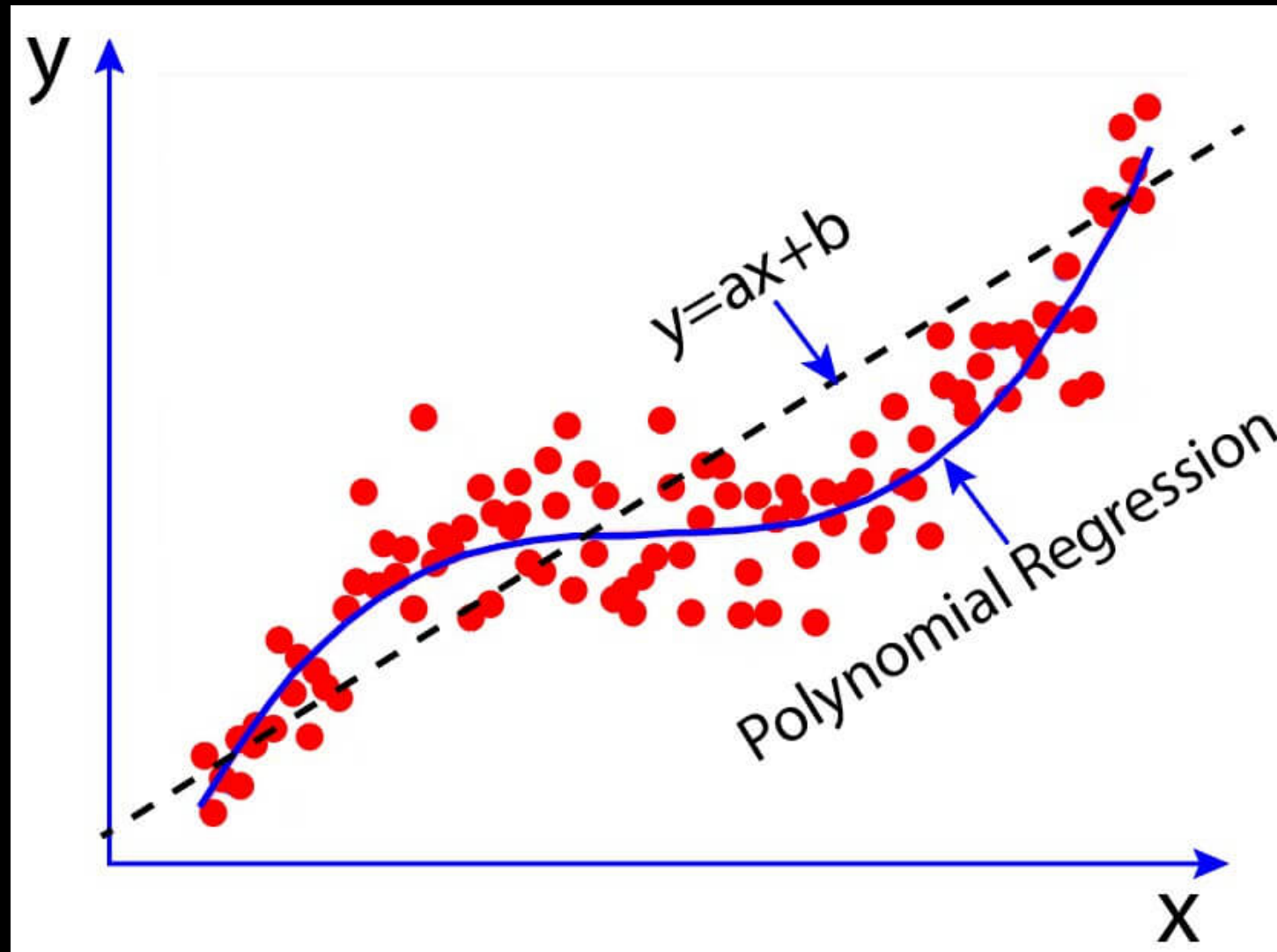
[29] ✓ 0.0s

... [[50.]]

Content

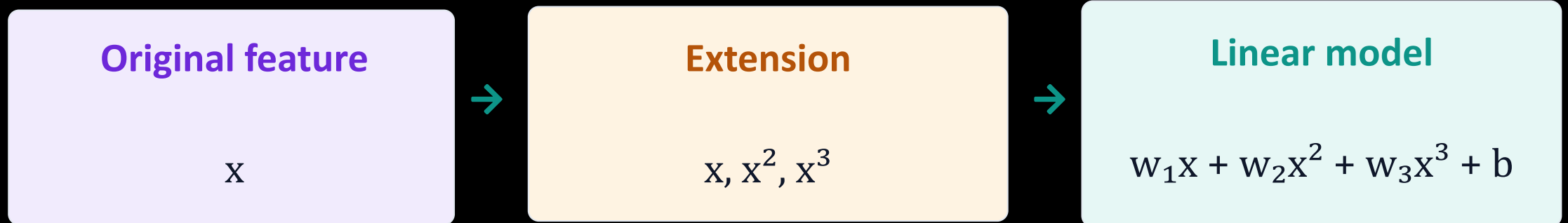
- ~~Linear Regression with Multi-Variables~~
- Polynomial Regression
- Setting Up Environments
- Implementation of Linear regression

Polynomial Regression



Polynomial Regression

Core trick: We construct new features from the input x —such as $x^2, x^3, \dots$ —and then train a standard linear regression on top of these expanded features.



Key point: The model is still linear with respect to the parameters w . The word "polynomial" refers only to the shape of the features.

Polynomial Regression

```
from sklearn.preprocessing import
PolynomialFeatures
from sklearn.linear_model import
LinearRegression
from sklearn.pipeline import make_pipeline

#
poly_model = make_pipeline(
    PolynomialFeatures(degree=2),
    LinearRegression()
)

poly_model.fit(X_train, y_train)
y_pred = poly_model.predict(X_test)

# PolynomialFeatures: x -> [1, x, x^2]
```

Polynomial Features

Generates polynomial features ($x^2, x^3, \dots$) from x .

make_pipeline

Combines the steps into a single model—convenient and error-free.

degree

The main hyperparameter—it must be chosen correctly!

Content

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